

Pricing under a Hidden Markov Regime by the Fast-Switching Expansion: The Reduced System and Its Asymptotics

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Abstract

A regime-switching model lets any parameter of a standard model take a different value in each state of a hidden Markov chain. For the affine and exponential-Lévy models in common use the pricing problem reduces to one linear system, $a' = (Q + \text{diag } g(t)) a$, with the chain's generator Q and a model-specific forcing g as its only ingredients. We expand its solution in the mean holding time of the chain to any order, with an initial layer for the starting regime, and identify the first correction as a Green–Kubo matrix from the group inverse of Q . The same system prices bonds, options on bonds and swaps, and equity options under stochastic rates once the Gil–Pelaez integrals are conditioned on the regime at expiry. Every numerical claim is checked against **QuantLib** in the limit where the regimes coincide and against an independent referee with switching on, and the implementation is public.

1 Introduction

Hamilton's regime-switching model [6] put a hidden finite-state Markov chain behind the parameters of a time series, and Naik [11] priced options with the volatility driven by such a chain. The pricing problem under a regime-switching model is a coupled system, one equation per regime, and for the affine and exponential-Lévy models in use the state dependence factors out, leaving a linear system of ordinary differential equations in which the chain and the model meet only through the generator and a forcing term. The reduction, and the asymptotic expansion of its solution when the chain is fast in the mean holding time, were carried out for the Ornstein–Uhlenbeck bond and the survival probability in the author's thesis [4]. The two-state reduction for the Vasicek bond appears again in Elliott and Mamon [5], and American options with switching in Buffington and Elliott [3].

Yin [16] expanded the coupled Black–Scholes equations in the holding time of a fast chain, with the averaged volatility at leading order and uniform error bounds, and Basu and Ghosh [2] corrected the risk-minimising option price for a fast and for a slow chain. The general theory of two-time-scale Markov chains, forward and backward equations without a potential term, is Yin and Zhang [18] and Yin and Nguyen [17]; Nguyen and Yin [12, 13] carried the same expansion, with a terminal layer and uniform error bounds, to the backward equation of a switching diffusion, where the regime enters through the diffusion operator rather than a potential. The parallel

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expansion for fast stochastic volatility, with a chain among the admissible drivers, is Sircar and Papanicolaou [14], and for a fast diffusion driving the volatility Khasminskii and Yin [8].

The reduced system has one form for every model in QuantLib [1]: a forcing $g_i(t)$ per regime. Its solution expands in the mean holding time to any order, with an initial layer, and the terms are the averaged model at order zero, a Green–Kubo matrix of the forcing’s fluctuations at order one, and a memory of the starting regime through the group inverse of the generator. The series is asymptotic. Where it fails, the reduced system is solved numerically instead, and the engine reports which.

The instruments a library must price follow from the same system: bonds, European and digital options, options on bonds and swaps, caps, Asian, barrier, American and Bermudan instruments, credit default swaps, equity options under stochastic rates. Two constructions in that list appear to be new. Gil–Pelaez integrals conditioned on the regime at expiry make Jamshidian’s decomposition [7] work regime by regime, and a discounted characteristic function prices equity options when the rate and the underlying share the chain.

The implementation, `regimelib`, follows QuantLib’s classes with a chain in front of the parameters. Every model reduces to QuantLib’s when the regimes coincide, and a suite of certificates checks that limit against QuantLib and the switching case against a numerical, grid or Monte Carlo referee. The tables below are written by a script in the repository.

2 The reduced system

Let y_t be a continuous-time Markov chain on $\{1, \dots, n\}$ with generator Q (rows sum to zero, off-diagonal entries nonnegative), stationary distribution π , and let x_t be the state of a model whose parameters depend on y_t . For a price of the form $u_i(t, x) = \mathbb{E}[\Phi(x_T) e^{-\int_t^T r_s ds} \mid x_t = x, y_t = i]$ the pricing equations are coupled through the generator,

$$\partial_t u_i + L_i u_i + \sum_j Q_{ij} u_j = 0, \quad u_i(T, \cdot) = \Phi, \quad (1)$$

with L_i the generator of x in regime i less the discounting. When the model is affine in x and the payoff exponential-affine, $u_i(t, x) = \exp(-B(\tau) \cdot x) a_i(\tau)$ with $\tau = T - t$, B solving the model’s Riccati equation (regime-free when only drift levels, volatilities, correlations or jump intensities switch), and a solving

$$a'(\tau) = (Q + \text{diag } g(\tau)) a(\tau), \quad a(0) = a_0, \quad (2)$$

where $a_0 = \mathbf{1}$ for a payoff paid in every regime and $a_0 = e_j$ for one paid only if $y_T = j$. Everything model-specific is in the forcing $g_i(\tau)$, and Table 1 lists it for the models in the library. Characteristic functions are the same statement with a complex payoff exponent: for the martingale log return X_T of an equity model, $\mathbb{E}[e^{iuX_T} \mid y_0 = i] = c(u) a_i(T)$ with the forcing evaluated at u .

Proposition 1 (Exact reductions). *For every model in Table 1 the coupled system (1) is solved by $\exp(-B(\tau) \cdot x) a(\tau)$ with a from (2). The reduction fails when a switched parameter multiplies an operator that does not commute with the rest of L_i : the CEV diffusion $\sigma_i^2 S^{2\beta} \partial_{SS}$ against the drift, or the Heston volatility of variance ξ_i against the mean reversion.*

The two-state case of the reduction, a pair of linear ODEs, is in Cotton [4] and again in Elliott and Mamon [5] for the Vasicek bond. One form covers the whole catalogue, including the

Model	switching parameters	forcing $g_i(\tau)$
Vasicek $dr = a(b - r)dt + \sigma dW$ with a terminal exponent e^{-cx_T}	b, σ	$-ab_i B + \frac{1}{2}\sigma_i^2 B^2, B = (1 - e^{-a\tau})/a$ same with $B_c = ce^{-a\tau} + (1 - e^{-a\tau})/a$
Vasicek with exponential jumps	b, σ, λ	$\dots + \lambda_i((1 + mB)^{-1} - 1)$
CIR $dr = k(\theta - r)dt + \sigma\sqrt{r}dW$	θ	$-k\theta_i B, B$ the CIR Riccati solution
Hull–White $r = x + \varphi(t)$	σ	$\frac{1}{2}\sigma_i^2 B^2, \varphi$ fitted with $\bar{\sigma}^2 = \pi \cdot \sigma^2$
G2++ $r = x + y + \varphi(t)$	σ, η, ρ	$\frac{1}{2}(\sigma_i^2 B_a^2 + \eta_i^2 B_b^2 + 2\rho_i \sigma_i \eta_i B_a B_b)$
Black–Scholes, log return	σ	$-\frac{1}{2}\sigma_i^2(u^2 + iu)$
Heston, long-run variance	θ	$\kappa\theta_i D(\tau; u), D$ the Heston Riccati solution
Merton, Bates	σ or θ, λ	$\dots + \lambda_i(\phi_J(u) - 1 - iu\bar{\kappa})$
Variance gamma	σ, ν, θ	$-\nu_i^{-1} \log(1 - iu\theta_i \nu_i + \frac{1}{2}\sigma_i^2 \nu_i u^2) + iu\omega_i$
Geometric Asian (Black–Scholes)	σ	$iu(\mu - \frac{1}{2}\sigma_i^2)\frac{\tau}{T} - \frac{1}{2}u^2\sigma_i^2\frac{\tau^2}{T^2}$
Intensity basket $\lambda^1 + \dots + \lambda^K$	each name's	sum of the names' forcings
Equity with rates (discounted c.f.)	both parts'	$g_i^{\text{rates}}(\tau; c = 1 - iw) + g_i^{\text{equity}}(w)$

Table 1: The forcing of the reduced system (2) for the models in `regimelib`. B, D are the regime-free Riccati solutions of the frozen model; the prefactor $\exp(-B \cdot x_0)$ or $\exp(Dv_0)$ is common to all regimes. The Asian forcing is in time to maturity, where the weight $(1 - t/T)$ of the log price becomes τ/T .

terminal exponents that bond options need and the complex forcings of characteristic functions. The models outside the catalogue are exactly those with a non-commuting switched operator, and they are priced at first order below.

3 The fast-switching expansion

Write $Q = Q_1/\varepsilon$ with $\varepsilon = n/(-\text{tr } Q)$ the mean holding time of the chain and Q_1 of unit scale, and expand $a = \sum_k \varepsilon^k a^{(k)}$. Substituting into (2) and matching powers gives at each order an equation $Q_1 w = f$, which has a solution only when $\pi \cdot f = 0$ (the Fredholm alternative for the singular matrix Q_1); the solvability conditions determine the projections onto $\mathbf{1}$, the particular solutions are given by the group inverse $Q_1^\#$ [10], and the initial condition $a(0) = a_0$, which the outer expansion cannot satisfy regime by regime, is met by an initial layer in the fast time τ/ε . The construction is standard singular perturbation [13, 18]; the potential term $\text{diag } g$ is what separates the reduced system from the backward equations treated there. The terms have a form that can be stated.

Theorem 2 (Expansion with initial layer). *Let g be smooth on $[0, T]$ and Q irreducible. For every N there are functions $a^{(0)}, \dots, a^{(N)}$ and layer terms $\ell^{(0)}, \dots, \ell^{(N)}$, each computable from $Q_1^\#, \pi$ and g by quadrature, such that*

$$a(\tau) = \sum_{k=0}^N \varepsilon^k (a^{(k)}(\tau) + \ell^{(k)}(\tau/\varepsilon)) + O(\varepsilon^{N+1})$$

uniformly on $[0, T]$, with $\ell^{(k)}$ decaying exponentially in τ/ε . The leading terms are

$$a_i^{(0)}(\tau) = \exp\left(\int_0^\tau \bar{g}\right), \quad \bar{g} = \pi \cdot g, \quad (3)$$

$$a_i^{(1)}(\tau) = a^{(0)}(\tau) \left(\int_0^\tau K(s) ds - m_i(\tau) \right), \quad (4)$$

$$K(s) = -\pi \cdot (\tilde{g}(s) \odot Q_1^\# \tilde{g}(s)), \quad m_i(\tau) = (Q_1^\# \tilde{g}(\tau))_i,$$

where $\tilde{g} = g - \bar{g}\mathbf{1}$ is the centred forcing and $\odot$ the entrywise product; $\ell^{(0)} = 0$ and the first layer is $\ell^{(1)}(s) = e^{Q_1 s} Q_1^\# \tilde{g}(0)$, which cancels $-m(0)$ at $\tau = 0$ and decays because $Q_1^\# \tilde{g}(0)$ has no component along $\mathbf{1}$ ($\pi Q_1^\# = 0$).

The order-zero term is the averaged model: the forcing replaced by its stationary average, which for Vasicek is the bond of the averaged mean level and averaged variance. The first correction has two parts. The integral of K is a Green–Kubo term: for a vector forcing $g = \sum_j \phi_j f_j$ with regime-dependent coefficients f_j it is the quadratic form $\sum_{jk} \phi_j \phi_k K_{jk}$ in the matrix

$$K_{jk} = \int_0^\infty \text{Cov}(f_j(y_0), f_k(y_t)) dt = -\pi \cdot (\tilde{f}_j \odot Q_1^\# \tilde{f}_k), \quad (5)$$

the integrated autocovariance of the parameters along the chain, the extra convexity the switching creates. Only its symmetric part enters a scalar forcing. The antisymmetric part, nonzero for a non-reversible chain, enters when the switched operators do not commute.

The memory term m_i is the group inverse applied to the centred forcing, evaluated in the starting regime. It distinguishes the price seen from regime i from the price seen from regime j , and it decays with the layer as the chain forgets its start. Higher orders follow by the same recursion, and the library computes them to any order, with the integrals done exactly when g is an exponential sum and by Chebyshev series otherwise.

The certificate `verify_theorem.py` in the repository evaluates the residual of the order-one approximation, layer included, for a three-regime chain with the Vasicek forcing and with a constant forcing: the supremum over $[0, T]$ divided by ε^2 is constant to within a few percent across $\varepsilon = 0.2, 0.1, 0.05, 0.025$, and without the layer the residual for the constant forcing is 0.043ε , so the layer is not optional.

Remark 3 (The series is asymptotic). The expansion parameter is not ε alone but $\varepsilon \|g\|$, and for a characteristic function g grows with the Fourier variable, $|g_i(u)| \sim \frac{1}{2}\sigma_i^2 u^2$ for Black–Scholes. At fixed ε the terms therefore stop decreasing beyond some order, and at fixed order they diverge beyond some frequency. The engine therefore reports the size of the last term kept and stops at the best truncation when asked. At Fourier nodes where the last term is not smaller than the one before it, the reduced system is solved numerically instead of expanded, and the price is a hybrid: the expansion where it converges, the numerical solution where it does not. The geometric Asian option, whose characteristic function decays slowly in u , needs the hybrid at moderate holding times; replacing the diverging nodes by the averaged value leaves an error of 10^{-3} there.

Proposition 4 (Two regimes, constant forcing). *For $n = 2$ and constant g the solution of (2) is exact: with $M = Q + \text{diag } g$ and eigenvalues $\mu_\pm = \frac{1}{2}(\text{tr } M \pm \sqrt{\text{tr}^2 M - 4 \det M})$, $a(\tau) = [e^{\mu_+ \tau}(M - \mu_- I) - e^{\mu_- \tau}(M - \mu_+ I)]\mathbf{1}/(\mu_+ - \mu_-)$. This is the closed-form characteristic function of every two-regime Black–Scholes, Merton and variance-gamma model, and the expansion of Theorem 2 is its expansion in ε .*

4 Instruments

Bonds and European options. A bond is $\exp(-B(T) \cdot x_0) a_i(T)$ with $a_0 = \mathbf{1}$; with $a_0 = e_j$ it is the price of the face value paid only if $y_T = j$, and the sum over j recovers the bond. European options use Lewis's formula [9] with the reduced characteristic function,

$$C = e^{-rT} \left[F - \frac{\sqrt{FK}}{\pi} \int_0^\infty \operatorname{Re} \left(e^{iuk} \phi(u - \frac{i}{2}) \right) \frac{du}{u^2 + \frac{1}{4}} \right], \quad \phi(z) = c(z) a_i(T; z), \quad (6)$$

$k = \log(F/K)$, with the frequency cutoff found from the averaged characteristic function, which every model exposes in closed form. Differentiating under the integral gives delta, gamma, theta and rho, and for stochastic-volatility models the vega in the initial variance; theta uses $\partial_T \phi$, which the reduced system supplies as $(Q + \operatorname{diag} g(T)) a(T)$. Digitals are Gil–Pelaez integrals of the same function.

Options on bonds, swaptions, caps. Under a Gaussian short rate the bond at expiry T for maturity S in regime j is $A_j e^{-bx_T}$ with $A_j = a_j(S-T)$ from the reduced system and $b = B(S-T)$, so the option pays $(A_j e^{-bx_T} - K)^+$ on the event $\{x_T < x_j^*\}$ with one crossing $x_j^* = \log(A_j/K)/b$ per regime. Each term is a discounted probability of the form $\mathbb{E}[e^{-\int_0^T r} e^{-cx_T} \mathbf{1}\{x_T < x_j^*\} \mathbf{1}\{y_T = j\}]$. Write $\Psi_j(c - iu)$ for the reduced system with the terminal exponent $c - iu$ and $a_0 = e_j$; Gil–Pelaez inversion gives

$$\mathbb{E}[e^{-\int_0^T r} e^{-cx_T} \mathbf{1}\{x_T < x_j^*\} \mathbf{1}\{y_T = j\}] = \frac{1}{2} \Psi_j(c) - \frac{1}{\pi} \int_0^\infty \operatorname{Im} \left(e^{-iux_j^*} \Psi_j(c - iu) \right) \frac{du}{u}. \quad (7)$$

The regime indicator at expiry is carried by the initial vector e_j and costs nothing beyond one solve per regime; the integrals are done on Kronrod panels with an error estimate and halved until it is met. A coupon bond $\sum_k c_k A_{kj} e^{-b_k x}$ is monotone in x in every regime, so Jamshidian's decomposition [7] holds regime by regime: one crossing x_j^* per regime, and the option is a sum of zero-coupon bond options with strikes $A_{kj} e^{-b_k x_j^*}$. A receiver swaption is the call on the coupon bond at par, a payer swaption the put, a caplet $(1 + \tau K)$ puts on the period-end bond struck at $1/(1 + \tau K)$. Under Hull–White the deterministic part of the discounting scales the cash flows; under G2++ the same construction applies to the zero-coupon bond with $z = B_a x_T + B_b y_T$ as the single variable and a two-factor terminal forcing.

Equity with stochastic rates. For an equity and a short rate driven by one chain with independent Brownian motions, the discounted characteristic function $\Psi(w) = \mathbb{E}[e^{-\int_0^T r} S_T^{iw}]$ factors, given the regime path, into the rates part $\mathbb{E}[e^{(iw-1)\int r}]$, the Gaussian terminal formula with weight $c = 1 - iw$, and the equity's martingale characteristic function, so the reduced system takes the sum of the two forcings (Table 1). Lewis's formula in discounted form, $C = S_0 e^{-qT} - \frac{\sqrt{K}}{\pi} \int_0^\infty \operatorname{Re} \left(K^{-iu} \Psi(u - \frac{i}{2}) \right) \frac{du}{u^2 + 1/4}$, prices the option with the discount and the return dependent through the regime. An equity–rate correlation is admitted for the Black–Scholes equity, where the exponent stays Gaussian given the path and the cross term $-ciw \rho \sigma_S \sigma_r B(\tau)$ joins the forcing. In the frozen limit these agree with QuantLib's analytic Black–Scholes–Hull–White and Heston–Hull–White engines to 10^{-8} and 10^{-6} relative.

Early exercise, barriers, baskets. American and barrier options, Bermudan swaptions and coupon-bond options are priced on the coupled system (1) directly, by Crank–Nicolson after Rannacher’s implicit start on a grid in the state with one block per regime, projecting onto the payoff at each step or exercise date; the exercise value of a swap at a Bermudan date is the coupon bond per regime from the reduced system, on the grid. Knock-outs truncate the grid at the barrier; knock-ins are the vanilla less the knock-out. Two-factor models (G2++, a switched vol-of-vol, equity with rates) use a tensor grid, uniform or concentrated by the sinh map of Tavella and Randall [15]. A basket of default intensities on one chain has joint survival equal to the bond of the summed forcing, so a first-to-default swap prices as a single-name swap on the basket, and the default correlation the common regime induces is read off the joint and marginal survivals.

5 Models outside the exact reduction

When a switched parameter multiplies an operator A_j that does not commute with the averaged generator $\bar{L}$, the pricing equation is $\partial_t u_i + (\bar{L} + \sum_j \tilde{f}_j(i) A_j) u_i + \sum_k Q_{ik} u_k = 0$ and the expansion above still applies to the operator-valued system, with the first correction now

$$u \approx e^{T\bar{L}} u_0 + \int_0^T e^{(T-s)\bar{L}} \left(\sum_{jk} K_{jk} A_j A_k \right) e^{s\bar{L}} u_0 ds - \sum_j m_j(i) A_j e^{T\bar{L}} u_0, \quad (8)$$

the Green–Kubo matrix (5) applied through Duhamel’s formula, and the memory term acting on the averaged solution. The antisymmetric part of K now contributes through the commutators $[A_j, A_k]$, which is how a non-reversible chain leaves a trace that no reversible chain can. The library evaluates (8) on a grid by an augmented linear system exponentiated once, reports the relative size of the correction, and estimates the neglected second-order term as its square; on the CEV model with a switched volatility the estimate tracks the actual error against the coupled system to within a factor of two across chain speeds, and on Heston with a switched volatility of variance the first-order price is within 10^{-4} – 7×10^{-4} of the coupled two-dimensional PDE while the correction itself is 0.3–3% of the price.

6 Parameter sensitivities

Because the chain enters the first-order price only through π , K and m , sensitivities to the model’s parameters are the chain rule through those numbers, and their derivatives are exact.

Proposition 5. *Let dQ be a perturbation of the generator with zero row sums. Then*

$$d\pi = -\pi dQ Q^\#, \quad dQ^\# = -Q^\# dQ Q^\# + \mathbf{1} \pi dQ (Q^\#)^2 + (Q^\#)^2 dQ \mathbf{1} \pi,$$

and K and m vary accordingly: $dK_{jk} = -[d\pi \cdot (\tilde{f}_j \odot Q^\# \tilde{f}_k) + \pi \cdot (d\tilde{f}_j \odot Q^\# \tilde{f}_k) + \pi \cdot (\tilde{f}_j \odot dQ^\# \tilde{f}_k + \tilde{f}_j \odot Q^\# d\tilde{f}_k)]$ with $d\tilde{f}_j = -(\tilde{f}_j \cdot d\pi) \mathbf{1}$. For a perturbation of a per-regime coefficient $f_j(i)$, K is bilinear and m linear in the forcing, so the derivatives are K and m evaluated on unit vectors.

With the first-order bond as a symbolic expression in $\pi \cdot \theta$, $\pi \cdot \sigma^2$, K and m , $\partial P / \partial \theta_i$, $\partial P / \partial \sigma_i$ and $\partial P / \partial Q_{ab}$ are formulas rather than bumps; Table 5 checks them against finite differences of the engine at the same order.

Instrument	QuantLib engine	regimelib	QuantLib	rel. diff.
Vasicek bond, $T = 5$	Vasicek::discountBond	0.83445782	0.83445782	0.0e+00
Hull–White bond call	HullWhite::discountBondOption	0.014645681	0.014645681	2.4e-15
G2++ bond call	G2::discountBondOption	0.016784068	0.016784068	3.1e-13
Hull–White payer swaption 2×5	JamshidianSwaptionEngine	0.0084708386	0.0084708405	2.2e-07
Hull–White Bermudan swaption	FdHullWhiteSwaptionEngine	0.013171556	0.013171543	9.7e-07
Hull–White cap, 4 caplets	AnalyticCapFloorEngine	0.013715712	0.013715712	3.3e-15
Black–Scholes put	AnalyticEuropeanEngine	11.504233	11.504233	8.7e-13
Heston put	AnalyticHestonEngine	9.2806468	9.2806468	1.1e-11
American put	FdBlackScholesVanillaEngine	11.793521	11.793912	3.3e-05
Down-and-out call	AnalyticBarrierEngine	9.5800708	9.5800332	3.9e-06
Geometric Asian call	AnalyticContinuousGeometric...	5.8327546	5.8327546	1.7e-12
Black–Scholes–Hull–White put, $\rho = 0.4$	AnalyticBSMHullWhiteEngine	14.522982	14.522982	1.3e-12
CDS fair spread, CIR intensity	MidPointCdsEngine	0.015712784	0.015714177	8.9e-05

Table 2: Frozen limit: two identical regimes on a chain with rates 3 and 5 per year, against QuantLib. Grid engines (American, Bermudan, barrier) carry the grid’s discretisation error; the characteristic-function engines agree to quadrature accuracy.

ε	order 0	1	2	3	4	6
0.050	7.2e-03	4.5e-05	1.1e-07	1.4e-07	1.9e-09	8.5e-11
0.100	1.5e-02	1.7e-04	9.9e-07	2.5e-06	1.9e-08	1.9e-08
0.200	3.1e-02	5.5e-04	1.5e-05	4.6e-05	3.1e-06	diverged
0.333	5.3e-02	5.9e-03	1.6e-04	diverged	diverged	diverged

Table 3: Relative error of the expansion at each order against the numerical solution of the reduced system, for a 2×5 payer swaption under a two-regime Vasicek model, by mean holding time ε of a symmetric two-state chain.

7 Numerical certificates

Table 2 is the frozen limit: two regimes with identical parameters, the chain still switching, priced by regimelib and by QuantLib’s engine for the same instrument. The characteristic-function engines agree with QuantLib to quadrature accuracy; the grid engines carry their discretisation error; the credit default swap differs by 9×10^{-5} , which is the rounding of QuantLib’s schedule to whole days.

Table 3 is the expansion with switching on: a payer swaption under a two-regime Vasicek model (mean levels 6% and 2%, volatilities 1.5% and 0.8%) at each order against the numerical solution of the reduced system, for four holding times. The error falls with the order until the asymptotic nature of the series shows, later for faster chains. The order at which it stops improving is the one the adaptive engine selects.

Table 4 gives timings on a laptop. For a bond, one evaluation of the series, the expansion is the fastest route. For a Fourier integral the numerical engine solves the reduced system for every node in a single batched call, and the series evaluated node by node at order four costs more; the expansion supplies the formula, the diagnostics and the greeks under the integral, not speed. The Monte Carlo referee is pure Python and priced per path, and is a referee rather than an engine.

Table 5 gives the parameter sensitivities.

Instrument (switching on)	expansion, order 4	numerical	grid / Monte Carlo
Vasicek bond	2.6 ms	7.1 ms	54890 ms (MC, 10^5 paths)
Vasicek swaption	4812 ms	250 ms	50 ms (grid)
Black–Scholes call / American put	355 ms	7 ms	43 ms (grid, American)
Heston call	12777 ms	6911 ms	–

Table 4: Wall-clock time per price with switching on (two regimes, rates 12 and 8), Python on a laptop.

Parameter	formula	finite difference	rel. diff.
$\partial P / \partial \theta_1$	-0.62010531	-0.62010531	1.6e-11
$\partial P / \partial \sigma_2$	0.018654226	0.018654226	4.1e-10
$\partial P / \partial Q_{12}$	0.0028187193	0.0028187193	3.0e-10
$\partial P / \partial Q_{32}$	-0.0027397038	-0.0027397038	1.2e-09

Table 5: Parameter sensitivities of the first-order Vasicek bond under a three-regime chain, from the derivative of the group inverse and from central differences of the engine at order one.

8 Reproducibility

The implementation is released under the MIT licence at <https://github.com/microprediction/regimelib>, on PyPI as `regimelib`, with documentation at <https://regimeswitching.org/python/>. A model in the all-orders tier is a statement of its forcing; a model whose reduction is not exact goes to the first-order tier with its diagnostics. Instruments accept `QuantLib`’s payoff, exercise and date objects, and the greeks are computed by differentiating the pricing formula.

Every certificate checks the frozen limit against `QuantLib` and the switching case against a referee that does not share the method: the numerical solution of (2) for the expansion, the coupled PDE on a grid for the first-order tier, Monte Carlo with exact regime paths for the grids. The frozen limit alone is not enough, and two blind spots fix the design. With all regimes equal, a forcing integrates to the same value in calendar time and in time to maturity, so the frozen limit cannot detect the direction of time in a time-dependent forcing; with switching on, the Asian option moves by 7% between the two. And a reversible chain has a symmetric Green–Kubo matrix, so only a non-reversible test chain separates the symmetric part of K from the whole in a cross term, a 10^{-4} effect in the first-order coupon-bond formula. The certificates therefore run with switching on and on non-reversible chains.

Closed forms are in `regimelib.symbolic`: the two-state Vasicek bond to second order, the first-order bond under any chain for Vasicek, CIR (integrals closed by the Riccati identity $\int B^2 = \frac{2}{\sigma^2}(T - kI_1 - B(T))$) and Vasicek with jumps, the exact two-state constant-forcing solution, and the parameter greeks above, each checked against the engine at the same order to round-off. The tables are written by `papers/regimelib/verify_tables.py` and the theorem’s certificate is `verify_theorem.py` beside it.

9 Conclusion

Theorem 2 is stated for a smooth forcing and an irreducible chain, and its proof is the singular-perturbation argument of Yin and Zhang [18]; the certificate checks the order-one case numerically

and not the general one. The first-order tier has no second-order term and no certificate beyond the error estimate against the coupled PDE. We leave to future work the second order of that tier, the equity–rate correlation for a Heston equity, and ports of the reference implementation to other languages under the parity contract of regimeswitching.org.

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